

PNG Atmospheric Integrity Pilot

From Atmospheric Evidence to Admissible Consequence

PARTIES	Crusaders Air Solution (CAS), Papua New Guinea x TA-14 Authority, USA
PHASE 1 OBJECT	Barlow Industries Compound 6-Mile Port Moresby Papua New Guinea
PRESENT DETERMINATION	SITE SELECTED BASELINE NOT FROZEN PRE-FREEZE HOLD

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Purpose. Institutional technical companion to the PNG Atmospheric Integrity Pilot showroom. It preserves the distinction between observation, governed record, continuity, admissibility, reliance, authority, standing, commit, execution, and outcome.

Evidence boundary. This reference is not certification, regulatory approval, funding commitment, government endorsement, implementation authorization, technical validation, or authority for an operational intervention.

Document Control & State Vocabulary

CONTROL	VALUE
Document	PNG Atmospheric Integrity Pilot - Technical Governance & Pre-Freeze Constitution Reference
Version	v2.0
Document status	Technical governance reference - PRE-FREEZE
Parties	Crusaders Air Solution (CAS) x TA-14 Authority
Phase 1 object	Barlow Industries Compound, 6-Mile, Port Moresby
English layer	Working technical baseline
Tok Pisin + Motu	Local CAS review; not treated as confirmed until returned
Current disposition	HOLD
Next evidence	CAS five-part confirmation / correction / UNKNOWN response
Continuity rule	APPEND - PRESERVE - DEEPEN; later states do not rewrite prior states

State Vocabulary

STATE	MEANING
ESTABLISHED	Supported by the presently preserved record.
SELECTED	Chosen as the candidate object; does not imply full constitution.
CANDIDATE	Proposed bounded basis awaiting required confirmation.
PRE-FREEZE	Constitution is underway; baseline is not yet frozen.
UNKNOWN / TBD	Material fact is not established and is preserved as such.
FROZEN	Mutually confirmed bounded baseline, only after explicit state transition.
HOLD	Current record does not authorize consequence now.

CHANGE CONTROL

Corrections and later evidence are appended with chronology and provenance. A later state may advance the record; it does not silently rewrite the state that existed before it.

1. Executive Technical Orientation

The pilot is designed around a simple but demanding distinction: monitoring can establish what was observed; governance determines what that evidence may legitimately support. A sensor alert, threshold crossing, inference, recommendation, or prior approval is not automatic authority for consequence.

CORE RULE

No admissible evidence. No admissible execution.

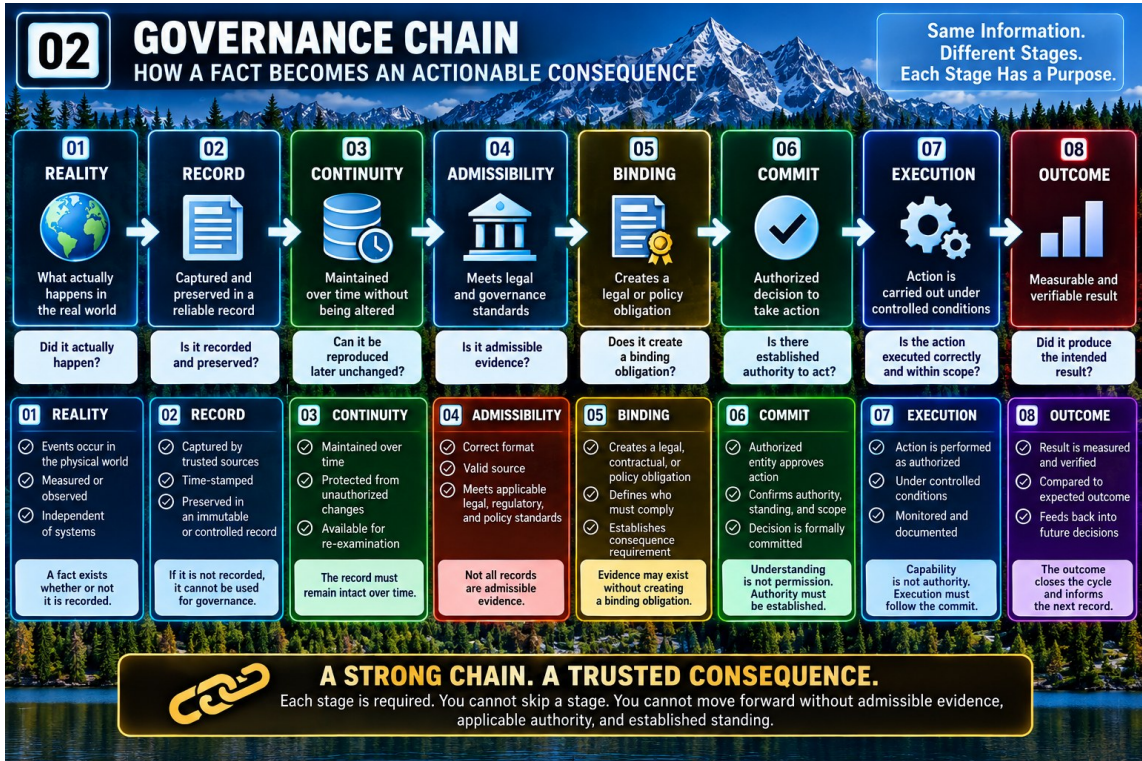
AUTHORITY RULE

Capability != Authority. Understanding != Permission. Commissioned once != authorized forever.

Current Phase 1 Control State

CONTROL	CURRENT RECORD	STATE
Phase 1 site	Barlow Industries Compound	SELECTED
Present site context	Administrative offices, primary warehouses, staff residential quarters	ESTABLISHED
English technical layer	Working technical baseline	ACTIVE
Tok Pisin + Motu	Under local CAS review	LOCAL REVIEW
Governance baseline	Seven bounded areas being constituted	PRE-FREEZE
Operational consequence	Not authorized by this record	HOLD
Next evidence	CAS five-part confirmation / correction / UNKNOWN	PENDING

2. Architecture Map - Evidence to Consequence



Teaching Visual 02 - explanatory representation; not site evidence.

STAGE	TECHNICAL QUESTION
REALITY	What actually existed?
RECORD	What was captured and preserved?
CONTINUITY	Has the basis remained materially continuous?
ADMISSIBILITY	May this record legitimately support this examination?
BINDING	What evidence, authority, and standing are bound to the exact proposition?
COMMIT	Has the bounded decision crossed the final authority boundary before action?
EXECUTION	What authorized action is performed?
OUTCOME	What became true in reality and must return to the record?

LOCAL AUTHORITY PRINCIPLE

Authority context may travel. Execution authority must be established locally for the exact consequence.

3. Pilot Ownership & Governance Boundary

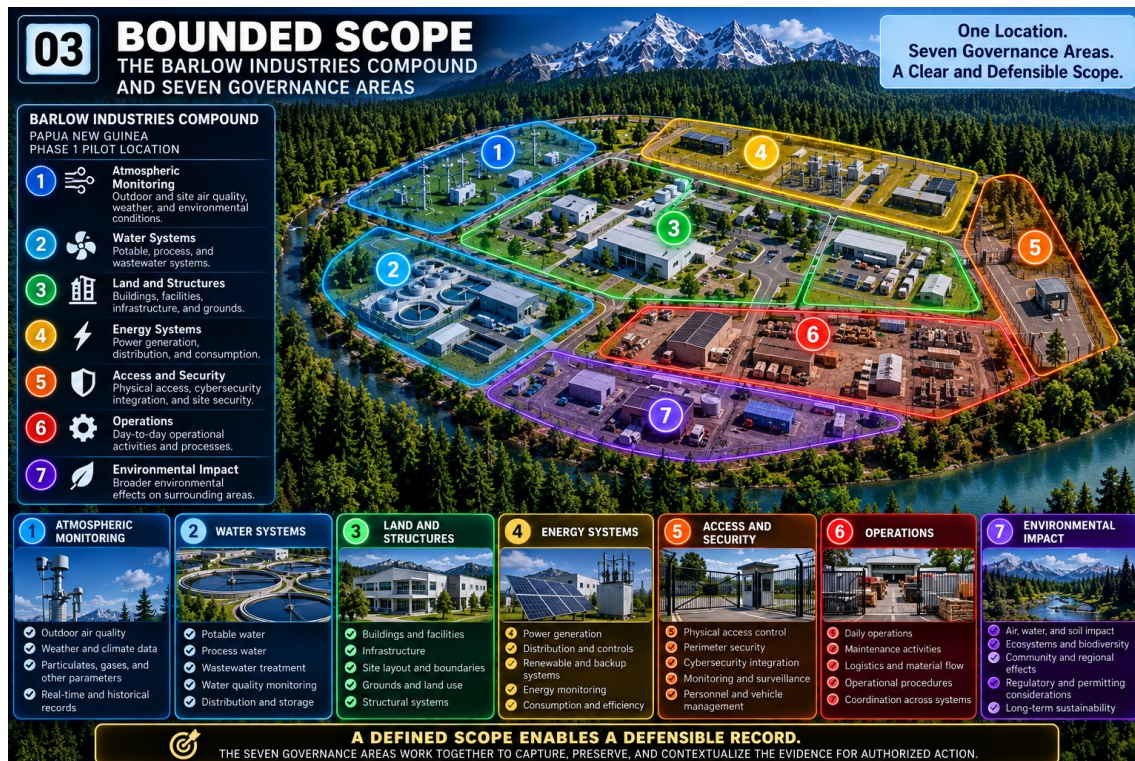


Teaching Visual 01 - conceptual illustration; not documentary photography of the Barlow Industries Compound.

Crusaders Air Solution retains the underlying pilot concept, site context, environmental/community objectives, and project authorship. TA-14 contributes governance architecture around Atmospheric Integrity Records (AIR), continuity, provenance, threshold separation, reliance, authority, controlled consequence, and changed-context re-examination.

Collaboration does not transfer execution authority by implication. A shared technical surface can preserve evidence and authority context while still requiring applicable authority to be established for the exact proposed consequence.

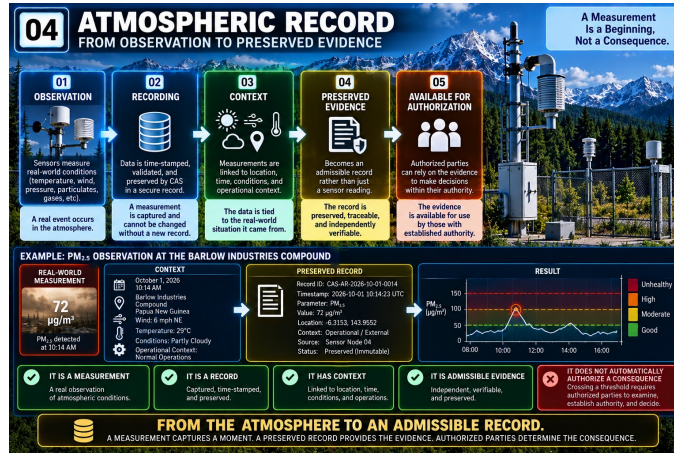
4. Bounded Phase 1 Scope



Teaching Visual 03 - conceptual scope diagram; not a surveyed site plan or documentary site image.

AREA	BOUNDED QUESTION
01 SITES	Physical Phase 1 boundary: included/excluded offices, warehouses, residential areas, monitoring locations, and relevant systems.
02 RECORDS	Minimum AIR content and provenance needed for later reliance.
03 THRESHOLDS	Environmental, regulatory, operational, or other thresholds; their source; and what each may legitimately support.
04 RELIANCE	Who or what may rely on each record, for what purpose, and under what conditions.
05 AUTHORITY	Who may qualify evidence, declare breaks, issue alerts, authorize reliance, or authorize a proposed consequence.
06 CONTINUITY	Material changes that require evidence, reliance basis, authority, or standing to be examined again.
07 CONSEQUENCE	Bounded actions evidence might eventually support and what must be established before execution.

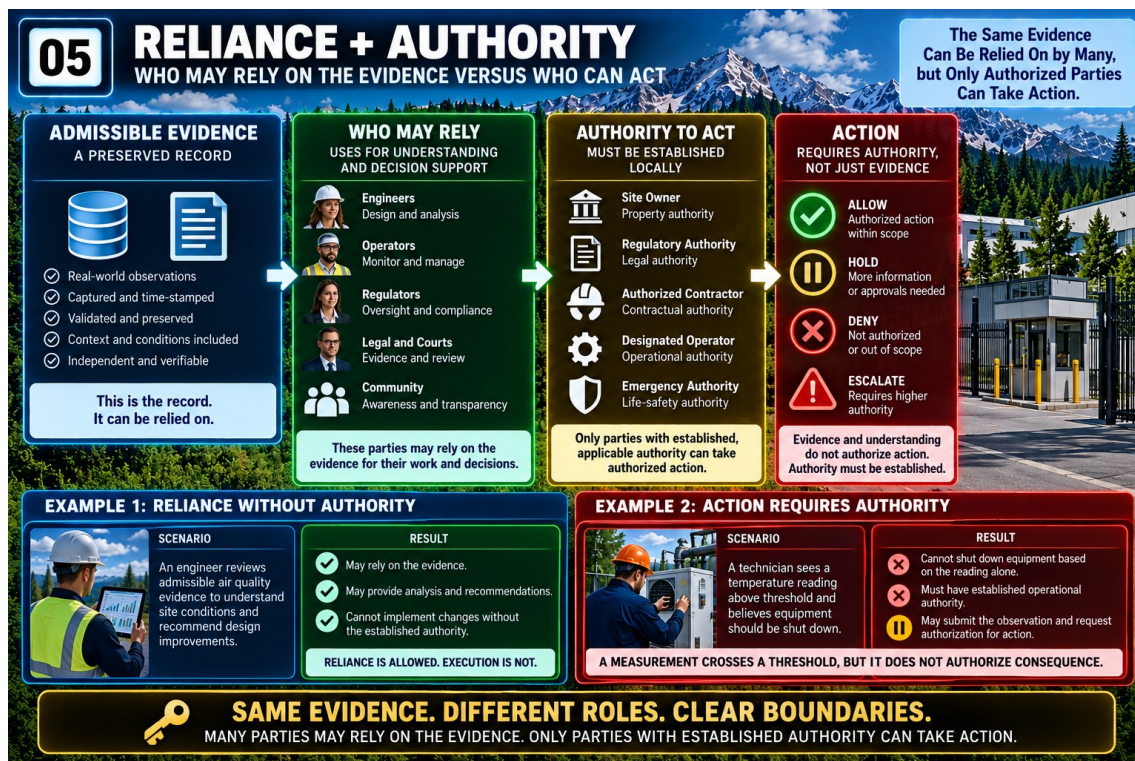
5. Atmospheric Integrity Record (AIR) - Minimum Technical Schema



Teaching Visual 04 - conceptual record model; not a live Barlow sensor record.

FIELD	PURPOSE	SOURCE	REQ.	CONTINUITY EFFECT
Timestamp	Chronology	Source/system time	YES	Break if absent/unreliable
Site + source ID	Attribution	Site/device identity	YES	Break if substituted/unresolved
Measured condition	Observation	Sensor/record	YES	Re-examine if definition changes
Device / sensor state	Integrity	Device state	YES	HOLD if material state unknown
Calibration / quality	Quality basis	Calibration/QA record	YES	Revalidate after material change
Provenance	Chain of evidence	Origin + transformations	YES	HOLD on provenance break
Relevant context	Interpretation boundary	Site/operational context	YES	Re-examine on material context change
Known limitations	Qualification	Examiner/source	YES	Preserve; limits reliance
Material change	Change detection	Event/record	WHEN P RESENT	Triggers re-examination
Continuity break	Standing	Governance determination	WHEN P RESENT	Prior standing cannot silently continue
Threshold state	Comparison	Applicable threshold source	WHEN RELIED ON	Does not authorize action
Reliance status	Permitted use	Governance determination	YES FOR ACTION	HOLD if not established
Qualification / escalation / revalidation	Disposition history	Governance record	AS APPLI CABLE	Append; never overwrite prior state

6. Reliance, Authority & Material Change



Teaching Visual 05 - explanatory representation; not evidence of actual authority holders at Barlow.

Technical validity, permitted reliance, and authority to cause consequence are governed separately. A material change may leave the historical record intact while changing present standing.

MATERIAL CHANGE RULE

The record remains preserved, but prior reliance or authority cannot silently continue through a changed condition. Revalidation establishes a new bounded basis; it does not rewrite the old one.

7. Material-Change Revalidation Matrix

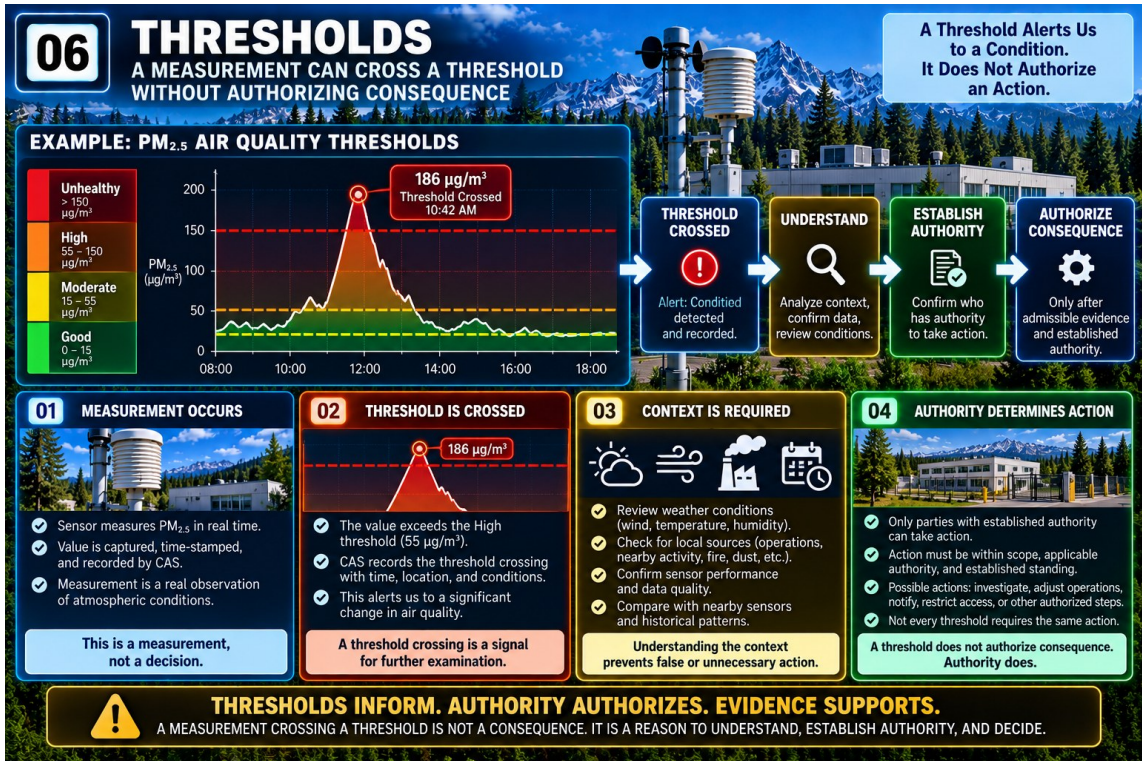
This matrix operationalizes the continuity rule without inventing site facts. It shows what the architecture requires when a material change is actually observed.

MATERIAL CHANGE	WHAT REMAINS PRESERVED	WHAT MAY LOSE STANDING	WHAT MUST BE RE-ESTABLISHED
Outage	Preserve prior record + outage event	Continuity / current standing	Restore evidence chain; re-examine gap
Sensor substitution	Preserve old and new identities	Continuity / comparability	Establish new source identity and quality basis
Recalibration	Preserve prior calibration state	Reliance / comparability	Establish changed calibration context
Data gap	Preserve gap explicitly	Continuity	Determine whether gap is material to reliance
Changed site/operational context	Preserve prior context	Standing / relevance	Re-examine reliance under new context
Provenance break	Preserve break and last known chain	Admissibility / reliance	Re-establish provenance or HOLD
Changed authority	Preserve prior authority state	Authority / standing	Establish current applicable authority

GOVERNING EFFECT

A material change does not erase history. It can change whether historical evidence may support a consequence NOW.

8. Threshold Separation

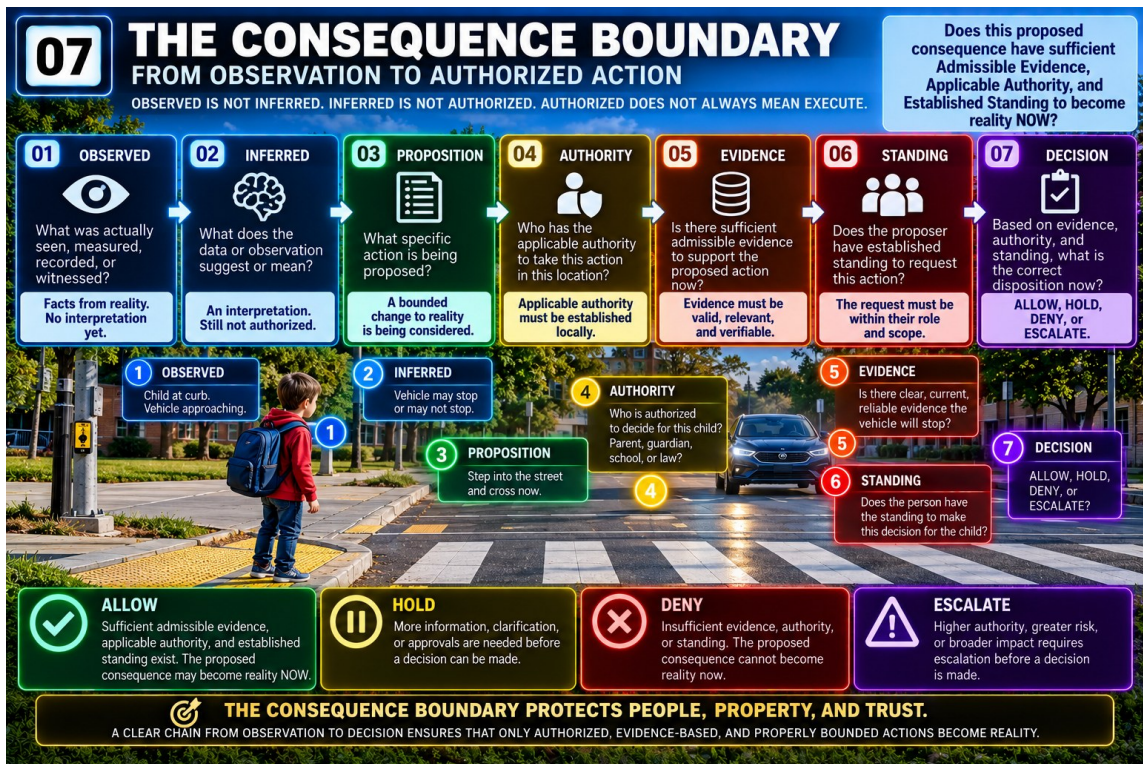


Teaching Visual 06 - conceptual threshold model; not a statement of actual Barlow thresholds.

CLASS	ROLE
ENVIRONMENTAL CONDITION	Measured atmospheric conditions.
REGULATORY / COMPLIANCE	Thresholds adopted or applied by the applicable authority.
HEALTH-OUTCOME EVIDENCE	May support correlation, escalation, or public-health reliance; not an atmospheric threshold itself.

A threshold crossing is an evidentiary condition to examine, not a command. Source, authority, intended reliance, continuity, and intervention authority still have to be established.

9. The Consequence Boundary

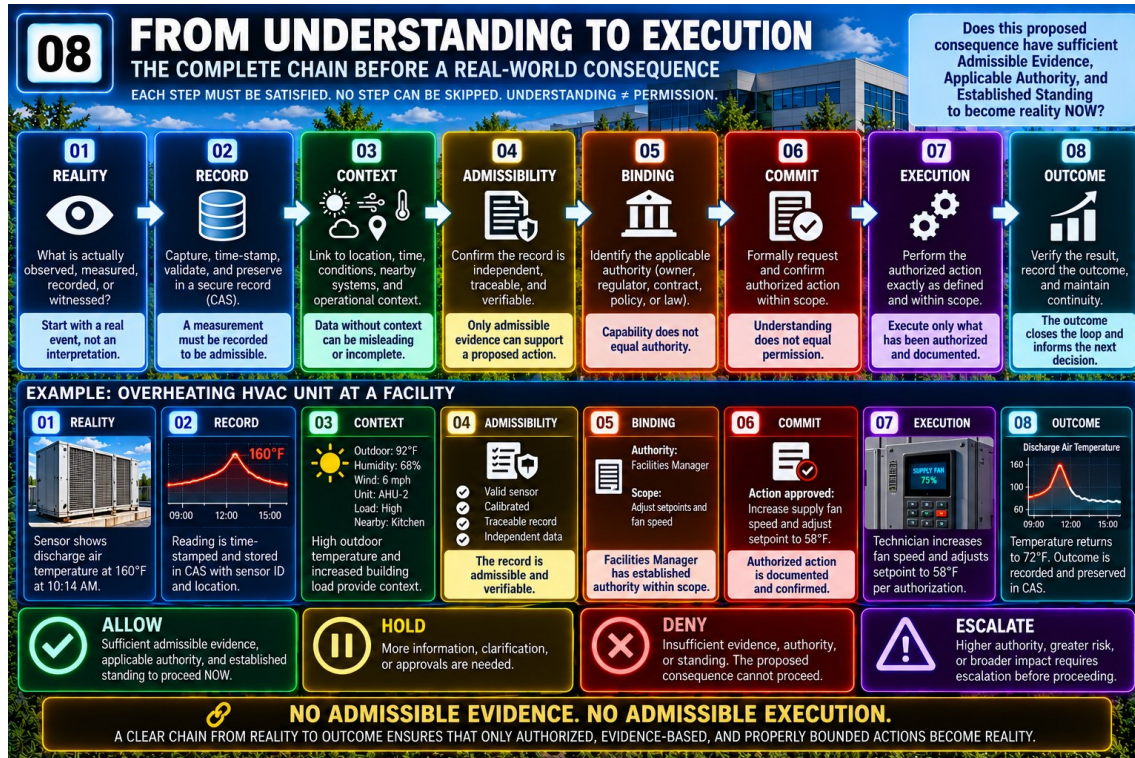


Teaching Visual 07 - conceptual consequence-boundary example; not an actual Barlow event.

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

DISPOSITION	USE WHEN
ALLOW	The exact proposed consequence is supported by sufficient admissible evidence, applicable authority, and established standing now.
HOLD	The record is not sufficient to permit consequence now but may advance through additional evidence, authority, standing, or revalidation.
DENY	The proposed consequence is not admissible under the established bounded basis.
ESCALATE	The proposition requires a different or higher authority, expertise, or governance path before determination.

10. From Understanding to Execution



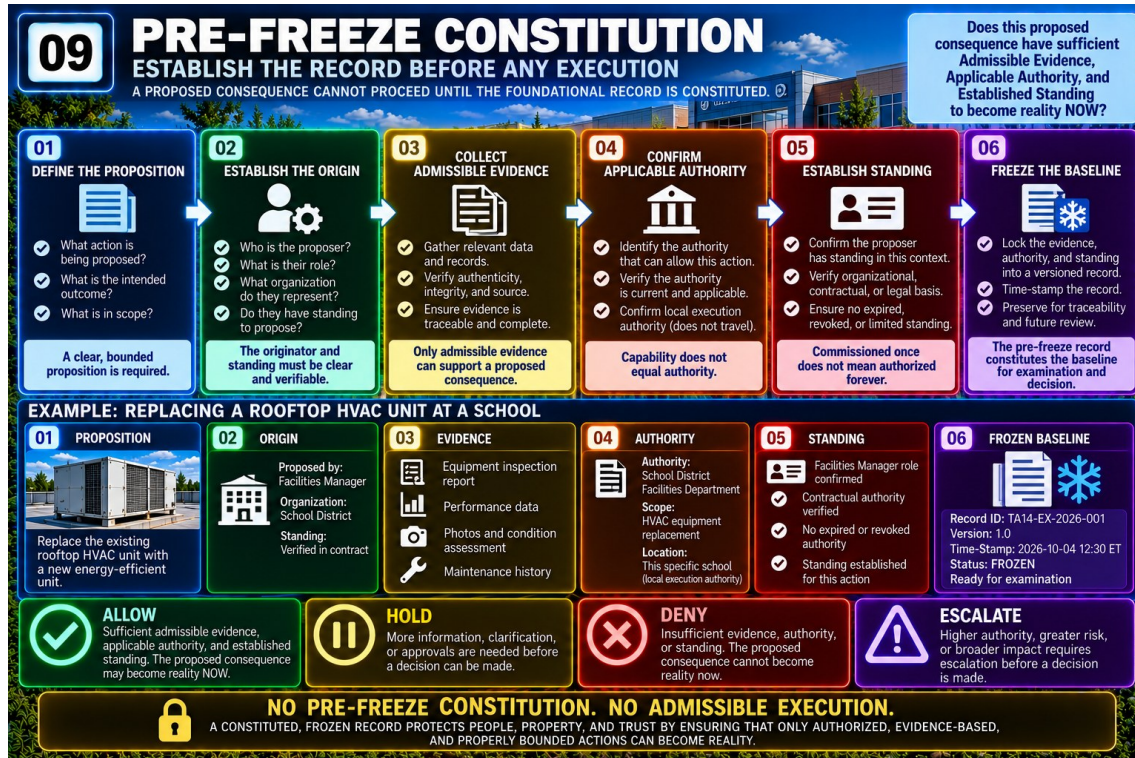
Teaching Visual 08 - conceptual workflow; not evidence of an actual Barlow execution event.

Selection of Barlow Industries Compound is meaningful institutional evidence. It establishes the candidate Phase 1 object and present site context. It does not establish all monitoring points, thresholds, authority holders, continuity conditions, or consequence permissions.

**CURRENT
DISTINCTION**

SITE SELECTED != FULL PHASE 1 GOVERNANCE SCOPE FROZEN

11. Pre-Freeze Constitution



Teaching Visual 09 - conceptual pre-freeze workflow. The PNG pilot is presently at PRE-FREEZE / HOLD; this visual does not assert that freeze has occurred.

A SITE SELECTED	Barlow Industries Compound is the candidate Phase 1 object.
B CONSTITUTE	Seven governance areas are made explicit.
C PRE-FREEZE CHECK	CAS returns confirmations, corrections, or UNKNOWN / TO BE DETERMINED.
D PRESERVE CORRECTIONS	Attributable changes are appended without erasing prior states.
E FREEZE	The mutually confirmed bounded baseline is designated FROZEN.
F EXAMINE LATER EVIDENCE	Later evidence and proposed consequences are examined against the preserved baseline and material changes.

12. CAS Pre-Freeze Confirmation Surface

The pilot does not need a new exploratory questionnaire. Five bounded confirmations are sufficient to advance the pre-freeze record. UNKNOWN / TO BE DETERMINED is an acceptable answer.

CAS may mark each item CONFIRMED, CORRECTED, UNKNOWN, or TBD and identify the evidence/source if available.

1. PHYSICAL BOUNDARY			Confirm included and excluded buildings, areas, or zones inside the Barlow Industries Compound.
STATUS	CAS RESPONSE / CORRECTION	EVIDENCE / SOURCE	
CONFIRMED / CORRECTED / UNKNOWN / TBD			

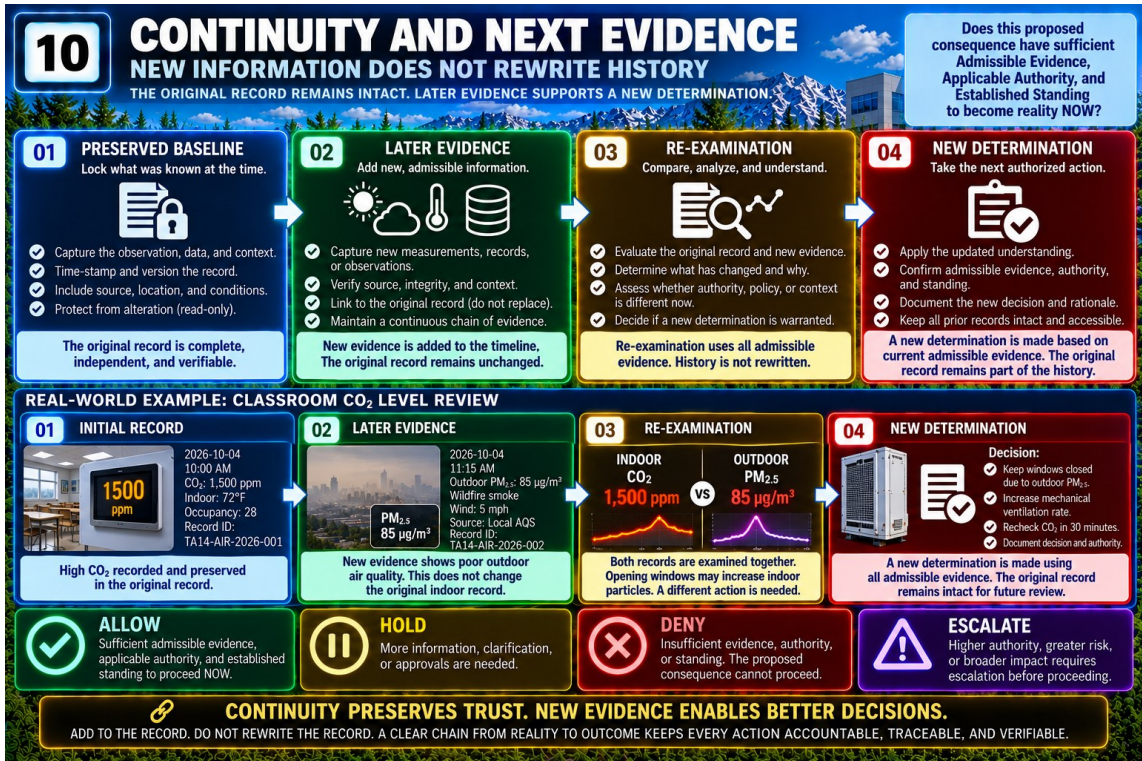
2. MONITORING POINTS / SYSTEMS			Identify intended monitoring points, devices, systems, or locations if presently known.
STATUS	CAS RESPONSE / CORRECTION	EVIDENCE / SOURCE	
CONFIRMED / CORRECTED / UNKNOWN / TBD			

3. THRESHOLDS			Identify existing environmental or operational thresholds intended for reliance and the source of each threshold.
STATUS	CAS RESPONSE / CORRECTION	EVIDENCE / SOURCE	
CONFIRMED / CORRECTED / UNKNOWN / TBD			

4. AUTHORITY			Identify who presently holds relevant environmental or operational authority at the compound.
STATUS	CAS RESPONSE / CORRECTION	EVIDENCE / SOURCE	
CONFIRMED / CORRECTED / UNKNOWN / TBD			

5. MATERIAL UNKNOWNNS			Identify anything material that is unknown, undecided, unavailable, or dependent on another party.
STATUS	CAS RESPONSE / CORRECTION	EVIDENCE / SOURCE	
CONFIRMED / CORRECTED / UNKNOWN / TBD			

13. Continuity & Next Evidence



Teaching Visual 10 - conceptual continuity model; not a record of future events.

APPEND - PRESERVE - DEEPEN. A later state may advance the record. It does not rewrite the state that existed before it. New evidence, corrections, determinations, and state transitions are appended with chronology and provenance.

NEXT STATE

TA-14 preserves the CAS response, returns the candidate baseline for the Pre-Freeze Constitution Check, incorporates attributable corrections without erasing prior states, and designates the baseline FROZEN only after mutual confirmation.

14. Institutional Continuity Record

What happened, when it happened, and what comes next. Earlier states remain visible as the institutional record deepens.

DATE / STATE	INSTITUTIONAL EVENT
24 Aug 2026	Exploratory material received from Crusaders.
24 Aug 2026	TA-14 returned Exploratory Governance Alignment Response v2; bounded Phase 1 scope recommended.
17 Sep 2026	Dedicated PNG governance-alignment showroom delivered for review.
18 Sep 2026	Crusaders confirmed review and intent to return feedback and Phase 1 object thoughts.
28 Sep 2026	CAS selected Barlow Industries Compound and described administrative offices, primary warehouses, and staff residential quarters.
28 Sep 2026	English technical layer prioritized; Tok Pisin and Motu remained under local review.
CURRENT	PRE-FREEZE CONSTITUTION - seven governance areas being constituted.
NEXT	CAS five-part confirmation / correction / UNKNOWN response.

CONTINUITY RULE

Record != present authority. Preservation != permission. Later evidence can advance standing without rewriting what was previously true or known.

15. Governing Architecture References

These references identify the TA-14 architectural seams used by this technical companion. They do not imply governmental adoption or external standard status.

ARCHITECTURE	ROLE IN THIS REFERENCE
AIR - Atmospheric Integrity Records	Preserves atmospheric observation with identity, chronology, provenance, quality context, limitations, and state.
EIG - Environmental Integrity Governance	Governs how environmental evidence may be relied upon and what consequences it may legitimately support.
AEA - Admissible Execution Architecture	Governs the transition from evidence and authority to real-world execution.
ACA - Admissible Computation Architecture	Relevant when computation or intelligence relies upon governed records or proposes consequence.
AFA - Admissible Federation Architecture	Relevant where authority context crosses system or organizational boundaries; execution authority remains local.
EABA - Execution Authority Boundary Architecture	Defines the commit boundary immediately before execution authority becomes real-world consequence.

FEDERATION RULE

Authority context can travel. Execution authority does not travel; it must be established locally for the exact proposed consequence.

16. Present Determination

PHASE 1 OBJECT	Barlow Industries Compound - selected candidate Phase 1 object.
BASELINE	NOT FROZEN. Seven bounded governance areas remain under constitution.
CURRENT DISPOSITION	HOLD AT PRE-FREEZE CONSTITUTION.
NEXT EVIDENCE	CAS five-part confirmation / correction / UNKNOWN response.

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Technical Evidence Boundary

This reference documents architecture and current preserved state. It does not establish certification, regulatory approval, government endorsement, funding commitment, implementation authorization, technical validation of a monitoring architecture, or authority for an operational intervention.

No admissible evidence. No admissible execution.

TA-14 Authority x Crusaders Air Solution | Technical Governance & Pre-Freeze Constitution Reference v2.0